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Interface features between $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ glassy cathode and $Li_7La_3Zr_2O_{12}$ solid electrolyte

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Glassy cathode deposition on solid electrolyte substrate applied to lithium middle-temperature all-solid-state batteries was investigated. The $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ (LBV) glassy cathode is crystallized on $Li_7La_3Zr_2O_{12}$ (LLZ) solid electrolyte substrate at various temperatures (650–850 °C) and holding times (0.5–5 min). The phase composition of the crystallized glass and vanadium oxidation state are determined by the XRD and XPS analysis. The different lithium vanadates form at all studied temperatures, and the boron-containing phases appear at temperatures greater than 750 °C. SEM investigation shows that the lithium borate-vanadate glass crystallized at 750 °C to create the optimal interface and close contact between the electrode and the electrolyte. Processing at lower temperatures does not allow for good contact, and processing at higher temperatures leads to the interaction between the cathode material and ceramic electrolyte. The optimal mode of $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ glassy cathode deposition on the substrate is found to be glass powder annealing for 0.5 min at 750 °C. This LBV|LLZ cell has the highest conductivity and the lowest polarization resistance on the cathode|electrolyte interface. The universality of this deposition mode is demonstrated using $Li_{3.65}Al_{0.05}Ge_{0.95}P_{0.2}O_4$ ceramic substrate as an example. Thus, the LBV glass can be used for the formation of an optimal interface with other lithium-ion solid electrolytes.

1. Introduction

All-solid-state batteries attract considerable scientific attention because their application can solve the major problems of energy storage. Such batteries have a number of advantages over commercially produced lithium-ion batteries, including increased safety, a wider operating temperature range, increased resistance to an aggressive atmosphere and high pressures, greater stability in the case of battery depressurization, and long lifetime [1–5]. Lithium lanthanum zirconate ($Li_7La_3Zr_2O_{12}$, LLZ) [6] and compounds based on it are the most promising electrolyte materials for all-solid-state batteries [7–13]. The main advantages of $Li_7La_3Zr_2O_{12}$ are electrochemical stability versus metallic lithium and a high lithium-ion conductivity of cubic modification at room temperature (~10^{-4} S cm^{-1}) [6]. Despite the fact that tetragonal modification of LLZ has a significantly lower value of conductivity (10^{-6} – 10^{-7} S cm^{-1} at 25 °C [14]), it possesses higher conductivity activation energy. So conductivity values of both modifications are comparable at 300 °C [15,16]. Moreover, the synthesis of tetragonal LLZ is considerably easier than cubic LLZ and does not need in addition of some stabilizing dopants. Therefore, LLZ of tetragonal modification can be used as the solid electrolyte in power sources operating at medium temperatures.

In the case of all-solid-state batteries, the main problem is to create an interface between solid electrodes and the electrolyte [17,18]. A low specific surface of the contact area leads to high polarization resistances in such power sources and to more rapid degradation processes on the electrodes (because of high overvoltage values). Hence, it is important to form a tight contact between electrodes and the solid electrolyte [19]. The interface modification at the anode|solid electrolyte interface was shown to be a useful technique for improving the cycle stability and the rate of a charge-discharge reaction. Some authors successfully used aluminum oxide (Al_2O_3) [17], Au [20], or Ge [21] deposition on LLZ substrate to develop a good contact between the metal anode and garnet electrolyte. Moreover, it has been shown that introducing Li_3PO_4 as an additive in garnet-type $Li_{6.5}La_3Zr_{1.5}Ta_{0.5}O_{12}$ improves

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the interfacial compatibility and suppresses Li-dendrite formation [22].

However, reduction of the interfacial resistance at the cathode|lithium garnet solid electrolyte interface remains a challenge [13,23]. For example, it was found in Ref. [24] that the high-temperature process of the $LiCoO_2$ and LLZ fusion induced the cross-diffusion of atoms and the formation of a less conductive phase at the interface. Some researchers use traditional cathode materials mixed with compounds having a low melting point to reduce the resistance at the cathode|lithium garnet interface (for example, Li_3BO_3 powder [25] or electrolytic salt [26]). In several works, some compounds were artificially delivered as a layer between cathode and electrolyte (Nb [27], liquid organic electrolyte [17] or gel membrane [21]). Presumably, these problems can be solved by using glassy cathodes or by adding conducting glass into the interface between the conventional cathode and solid electrolyte [28–31].

The glassy materials cast on solid electrolyte substrate, from the melt, can fill the free space (pores) between electrolyte grains, thereby creating a good and tight contact between them. It should be noted that glasses can partially or fully crystallize after casting of the glass melt on a ceramic substrate due to insufficient cooling rates. In the case of based-on-vanadium oxide glasses, the formation of lithium vanadates, which are the most attractive cathode materials, near the electrolyte surface is expected. Such glasses possess not only lithium-ionic conductivity but also to have high values of electronic conductivity (mixed conductivity) [32–36]. This allows such glasses (or their crystal analogs or glass-crystal composites) to be applied as the cathode materials for all-solid-state batteries. Vanadium oxides were extensively studied as cathode materials for batteries due to their attractive characteristics, such as high specific energy, good rate capacity, and stability during cycling [37–39]. The cathode ceramic materials based on the vanadium oxides have already been used, however, in a traditional way with carbon additive and polymeric binder [39]. The polymeric binder decomposes at a temperature above 100 °C, while the partially or completely crystallized glass can be used in the operating range of the middle-temperature power sources, which are the final objective of our investigation.

It has been shown that the lithium boron-vanadate glass crystallized on the solid electrolyte substrate based on LLZ can work as the cathode [40]. At 280 °C, the electrochemical cell was able to be charged up to 3.33 V. However, the optimal mode of cathode glass deposition and obtained interface properties were not investigated. The glass composition $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ (LBV) was shown in Ref. [41] to possess the highest total conductivity (ionic + electronic) in the $Li_2O–V_2O_5–B_2O_3$ system ($7.4 \cdot 10^{-6}$ S cm^{-1} at room temperature). In the course of the work, it is expected that the glass would crystallize partially under certain conditions and, even with a change in its composition, it would have high conductivity values. Therefore, to develop the optimal mode of the glassy cathode deposition, this work aims to determine the phase composition and conductivity of $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ glass crystallized under various conditions (different temperatures and holding time) on the tetragonal $Li_7La_3Zr_2O_{12}$ substrate.

2. Experimental

$Li_7La_3Zr_2O_{12}$ of tetragonal modification and $Li_{3.65}Al_{0.05}Ge_{0.95}P_{0.2}O_4$ (LAGP) were obtained by the solid-phase synthesis, which is described in Refs. [42,43]. Ceramic samples were obtained in the form of cylinders with the diameter of 15 mm and the thickness of ~1 mm; the relative density of the LLZ samples was ~75% that of the theoretical sample (5.1 g cm^{-3} [6]).

To obtain $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ glass lithium carbonate (Li_2CO_3), boric acid (H_3BO_3) and vanadium oxide (V_2O_5) were used as the starting materials. Appropriate amounts of the starting materials were taken in a stoichiometric relation and mixed thoroughly. The mixture was heated in a platinum crucible for 5 h at 1150 °C. The samples were obtained by the conventional quenching method and subsequently annealed at a temperature below the glass transition temperature (T_g) [41]. Then, the glass was ground in an agate mortar to produce a powder with a particle size ~5–10 μm.

The LBV glass powder (~15 mg) was applied to the solid electrolyte and evenly distributed over the sample surface with area ~176.6 mm^2. The thickness of the applied glass was ~0.1 mm. The obtained samples were placed in a furnace (LOIP, Russia) and kept at various temperatures (650–850 °C) for varying times (0.5–5 min).

X-ray diffraction analysis (XRD) was used to confirm the amorphous nature of the glass and to identify crystalline phases obtained after holding. XRD was performed with a Rigaku D-MAX-2200 V diffractometer with a vertical goniometer at Cu K_α-radiation and $2\theta = 10–55°$. The identification of compounds was carried out using a PDF-2 database (Release 2008 RDB 2.0804). The obtained data were processed using the FULLPROF program.

Impedance measurements of Ag|LBV|LLZ|GaAg|Ag were conducted in the air atmosphere using an immittance meter E7-25 (MNIPI, Belarus) in the frequency range of 0.025–1000 kHz in a two-probe cell with silver electrodes at temperatures from 110 to 330 °C. The LBV electrode completely covered the ceramic sample surface on one side; a gallium-silver paste (GaAg) or Pt were used as the electrodes on the other side. To check the reproducibility of the results, conductivity measurements were performed on several sets of samples.

The polarization resistance of a three-probe Ni|LBV|LLZ|GaAg|Ni cell was studied using a set of electrochemical techniques: chronopotentiometry, pulse chronopotentiometry, voltammetry, and electrochemical impedance. The LBV working electrode completely covered the ceramic sample surface on the one side, on the other side, there were two electrodes (GaAg) separated from each other, one of them was the current (counter) electrode, the other was potential (reference). Current collectors were made of nickel. The current was applied between working (LBV) and counter (GaAg) electrodes, while the voltage was received between working and reference (GaAg) electrodes. All of the measurements in the asymmetric cells were performed with LBV glass as the working electrode. Thus, the change in the counter electrode potential associated with the lithium intercalation or polarization at the counter electrode|electrolyte interface was excluded from the obtained data. All determined changes in the potential during the measurements correspond to the processes occurring on the working electrode. We used a P-5X galvanostat (Elins, Russia) with a current resolution of 10 nA, internal resistance of 10^{11} Ohm, and a pulse recording time of 2 μs. P-5X was equipped with an impedance module (FRA 24). The electrochemical impedance studies were performed in the 0.001–500 kHz frequency range with a perturbation voltage of ±25 mV. All of the measurements were carried out under cooling at the temperatures of 160, 210, 250, 290, and 330 °C. Before measurements, samples were held at each temperature for 2 h to achieve temperature equilibrium.

A chronopotentiometric study of the three-probe Ni|LBV|LLZ|GaAg|Ni cell was carried out at current density values of 5.4 μA cm^{-2}, 27 μA cm^{-2}, and 54 μA cm^{-2}. The pulsed method was carried out with a pulsed current of 20 μA and a pulse length of 20 μs; the data was processed by extrapolation to the current switching point. Voltammetry studies were proceeded by developing the implied potential in the range of -500–500 mV with a speed of 20, 2, and 0.2 mV s^{-1}. The processing of the obtained curves included the extraction of the cell resistance, determined by the pulse method, from the whole range of obtained data with the following transform to the dependency type of the overvoltage versus current density. The obtained polarization dependences were extrapolated to the zero scanning rate with the determination of the final polarization curve.

X-ray photoelectron spectra were obtained using a K-Alpha + XPS system (Thermo Fisher Scientific, USA) with monochromatic Al Kα radiation (hν = 1486 eV). X-ray photoelectron spectroscopy (XPS) analysis allowed us to determine the oxidation state of an element on the material surface (up to 10 nm). The analyzed area diameter was 400 μm. The binding energy scale was corrected using C1s peak (284.8 eV).

The morphology of the sample cross-section was investigated using a scanning electron microscope (Merlin, Carl Zeiss, Germany).

3. Results and discussion

1. Phase composition of $30Li_2O\cdot47.5V_2O_5\cdot22.5B_2O_3$ glass crystallized under different conditions

The XRD analysis revealed that the tetragonal modification of $Li_7La_3Zr_2O_{12}$ with *I4₁/acd* space group was obtained as the result of solid-state synthesis (Fig. 1a). The diffraction pattern of the initial $30Li_2O\cdot47.5V_2O_5\cdot22.5B_2O_3$ glass is a halo. This glass has the following characteristic temperatures: transition temperature, 210 °C; crystallization temperatures, 223 and 299 °C; melting temperature, 536 °C [41]. The LBV powder was crystallized on a $Li_7La_3Zr_2O_{12}$ electrolyte at various temperatures and holding times. When the temperature was 650 °C and the holding time was less than 3 min, the glass remained black, had a friable surface, and did not form the tight contact with the electrolyte. Holding time decreased as the temperature was increased; therefore, holding for 1.5 and 0.5 min was sufficient at 700 °C and at higher temperatures, respectively. The glass was deposited well and had a black color after holding at 750 °C; when temperature and holding time were increased, the melting glass spread and its color changed from black to brown (Fig. 1b). The color change might be due to the interaction of the LBV glass with the solid electrolyte substrate or the precipitation of the boron-containing phases upon devitrification of glass.

The XRD analysis was used to determine the phase composition of the cathode material on the LLZ substrate. The crystalline phases of LBV coatings obtained under different conditions are listed in Table 1. Depending on the chosen temperature mode, partial or complete crystallization of LBV glass with various phase formation takes place when the glassy material is deposited on LLZ substrate. LiV_3O_8 and LiV_2O_5 crystalline phases are formed after holding for

Table 1
Phase composition of the $30Li_2O\cdot47.5V_2O_3\cdot22.5B_2O_3$ glass, which was crystallized under various conditions on the $Li_7La_3Zr_2O_{12}$ solid electrolyte substrate.

Crystallization conditions	Phase composition
3 min 650 °C	LiV_3O_8 and LiV_2O_5
4 min 650 °C	$LiVO_3$, LiV_2O_5 and LiV_3O_8
5 min 650 °C	$LiVO_3$ and LiV_2O_5
1.5 min 700 °C	$LiVO_3$ and LiV_2O_5
2 min 700 °C	LiV_2O_5 and $LiVO_3$
3 min 700 °C	$LiVO_3$, Li_3VO_4 and LiV_2O_5
0.5 min 750 °C	LiV_2O_5 and LiV_3O_8
1 min 750 °C	LiV_2O_5 and $LiVO_3$
2 min 750 °C	Li_3VO_4 and $LiVO_3$
0.5 min 800 °C	$LiVO_3$, LiV_2O_5, Li_3VO_4 and B_2O_3
1 min 800 °C	Li_3VO_4, B_2O_3, $LiVO_3$, LiV_2O_5 and $Li_2B_4O_7$
1.5 min 800 °C	B_2O_3, $LiVO_3$, $Li_2B_4O_7$, LiV_2O_5 and Li_3VO_4
0.5 min 850 °C	B_2O_3, Li_3VO_4, LiV_2O_5, $Li_2B_4O_7$ and $LiVO_3$
1 min 850 °C	B_2O_3, $Li_2B_4O_7$, Li_3VO_4, LiV_2O_5 and $LiVO_3$
1.5 min 850 °C	B_2O_3, LiV_2O_5 and $Li_2B_4O_7$

3 min at 650 °C. When the holding time was increased to 4 min, LiV_3O_8, LiV_2O_5, and $LiVO_3$ crystalline phases were observed. After holding for 5 min, $LiVO_3$ became a predominant phase. LiV_2O_5 and $LiVO_3$ are formed at 700 °C and holding times of 1.5 and 2 min. The further increase in the exposure time resulted in the formation of an additional Li_3VO_4 phase.

The glass deposition for 30 s at 750 °C lead to the formation of LiV_2O_5 and LiV_3O_8 crystalline phases, as well as during holding for 3 min at 650 °C (Fig. 2). LiV_2O_5 and $LiVO_3$ phases are formed after holding at 750 °C for 1 min. This XRD pattern is identical to that of the cathode surface obtained after holding for 1.5 min at 700 °C. When holding time was increased to 2 min, the main Li_3VO_4 phase was formed. In the case of glass deposited at 700 °C, this phase is additional. Boron-containing $Li_2B_4O_7$ and B_2O_3 phases appear after annealing at 800 °C. Their fraction increases with increasing holding time. $LiVO_3$, LiV_2O_5, and Li_3VO_4 are the main phases of holding for 0.5 and 1 min. The formation of boron oxide occurs after annealing for 1.5 min at 800 °C. Boron and vanadium-containing phases of the same composition are also formed at the temperature of 850 °C, but there is a small redistribution in their concentration. Thus, it can be assumed that when holding temperature is less than 800 °C, the glass is partially crystallized and boron remains in amorphous part.

Since vanadium has various oxidation states (from V^{5+} to V^0), its oxidation state in the LBV cathode material, crystallized at various temperatures at minimal holding time, was determined by XPS analysis. The characteristic spectrum of vanadium consists of two spin-orbit splitting components, $2p_{3/2}$ (~517 eV) and $2p_{1/2}$ (~524.5 eV). The V2p spectrum approximation was carried out

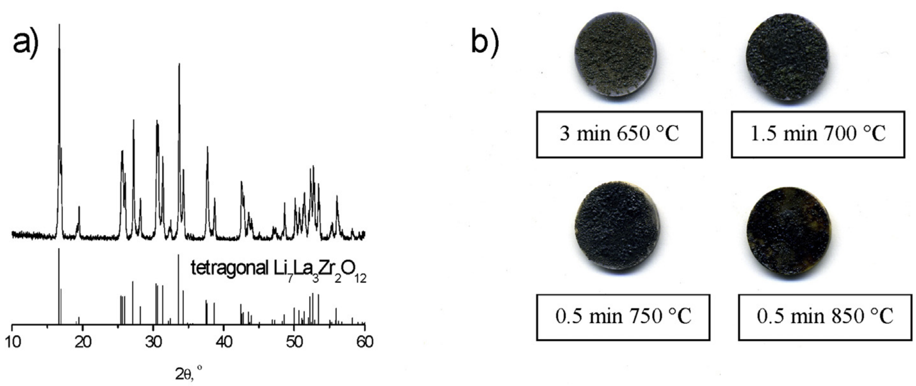

Fig. 1. XRD pattern of the synthesized tetragonal $Li_7La_3Zr_2O_{12}$ (a) and digital images of the LBV coated solid electrolyte (b).

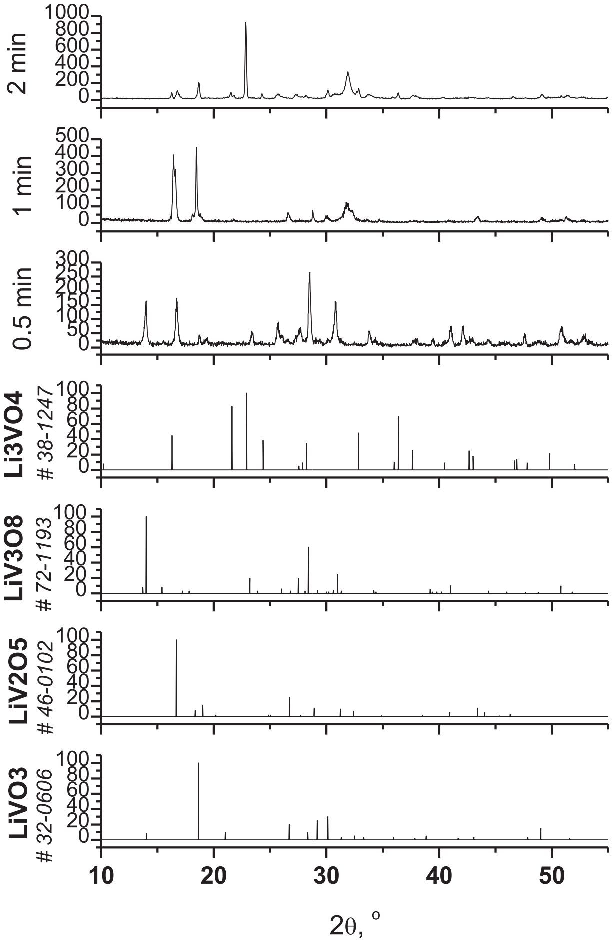

Fig. 2. XRD patterns of the crystallized LBV glass on the LLZ substrate for the various holding times at 750 °C.

using Shirley background subtraction, taking into account the O1s spectrum lines [44]. The ratio of V^{5+} and V^{4+} in the studied samples is given in Table 2. Two vanadium oxidation states, 5^+ (517 eV) and 4^+ (515.7 eV), were separated in the samples obtained at 650 and 750 °C (Fig. 3a and c). According to XRD analysis data, LBV glass is crystallized with the LiV_2O_5 and LiV_3O_8 formation under the mentioned conditions (Table 1). It is known that these compounds are vanadium bronzes, in which vanadium can be in two oxidation states (4^+ and 5^+) [45]. These compounds have a wide range of compositions and can easily operate as cathodes [45,46]. The percentage of V^{4+} ions in the sample obtained at 750 °C is higher than that obtained at 650 °C, but it does not exceed 10%. The vanadium is in the oxidation state of 5^+ in LBV glass crystallized at 700, 800, and 850 °C (Fig. 3b and d, Table 2). According to the XRD analysis, $LiVO_3$ and Li_3VO_4 phases are dominated by these conditions. According to [46], these phases contain vanadium ions only in the 5^+ oxidation

Table 2
The content of vanadium in different oxidation states in LBV, crystallized under various conditions.

Peak	Binding energy, eV	Atomic %				
		650 °C 3 min	700 °C 1.5 min	750 °C 0.5 min	800 °C 0.5 min	850 °C 0.5 min
V^{5+} 2p3/2	517.05	96.2	100	92.6	100	100
V^{4+} 2p3/2	515.70	3.8	0.00	7.4	0.00	0.00

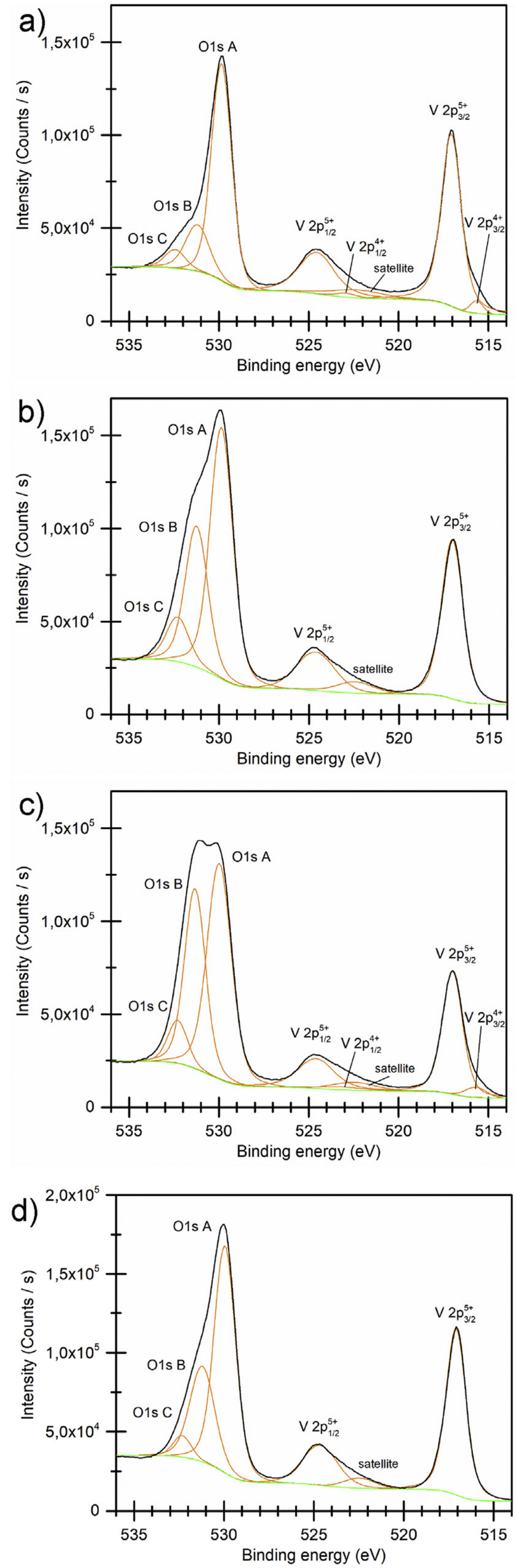

Fig. 3. V 2p and O 1s core-level XPS spectra of LBV at 650 (a), 700 (b), 750 (c), and 800 °C (d).

state and are attributed to the electrochemically irreversible compounds. Thus, the $30Li_2O\cdot47.5V_2O_5\cdot22.5B_2O_3$ glass deposition for 0.5 min at 750 °C leads to the cathode formation with optimal phase content for all-solid-state battery creation.

2. The electrochemical study of LLZ|LBV interface

The electrical resistance of the $Li_7La_3Zr_2O_{12}$ solid electrolyte and Ag|LBV|LLZ|GaAg|Ag cells, obtained at different temperatures and time conditions of glass deposition, were measured by the complex impedance method. Typical impedance plots of Ag|LBV|LLZ|GaAg|Ag cells obtained with the minimum holding time at different temperatures (650–850 °C) are shown in Fig. 4. The total resistance of the studied samples includes at least three components: the electrolyte resistance, the phase interface resistance between the electrode and electrolyte, and the resistance of the electrode (crystallized glass). Impedance spectra have a different shape for samples obtained under various conditions of the glass crystallization. For some samples, this consists of only one semicircle emerging from zero, which means that it contains all three contributions mentioned above. For others, one semicircle is also visible, but it does not come out of zero. It was found that the resistance value between zero and the point of the semicircle beginning refers to the resistance of the $Li_7La_3Zr_2O_{12}$ electrolyte since these values are in a good agreement with the values measured separately for the solid electrolyte. The semicircle contains the interface and the electrode resistances [47]. Therefore, it is extremely difficult to separate the contribution of each resistance component from the total; hence, the total resistance of the samples was determined from the real part of the impedance Z′ value at the minimum between the high-frequency arch and the low-frequency tail.

Table 3 indicates the total conductivity of the $Li_7La_3Zr_2O_{12}$ solid electrolyte and Ag|LBV|LLZ|GaAg|Ag cells, in which the glass deposition was carried out at various temperature and minimal holding time. Samples obtained at a minimum holding time of glass deposition (3 min at 650 °C, 1.5 min at 700 °C and 0.5 min at 750, 800, and 850 °C) were chosen because they had the highest conductivity at each holding temperature. Moreover, the samples obtained for 0.5 min at 750 °C possessed the highest conductivity in the studied temperature range, 1×10^{-6} and 4×10^{-4} S cm^{-1} at 140 and 330 °C, respectively. Since the conductivity values for samples obtained at temperatures <750 °C are lower (2×10^{-7} S cm^{-1} at 140 °C), it can be suggested that the good contact between the solid electrolyte and the electrode has not been achieved. A decrease in the electrical conductivity of the samples obtained at temperatures

Table 3
The total conductivity of LLZ (σ_{LLZ}) and samples with LBV glass crystallized on the LLZ substrate under various conditions ($\sigma_{LBV|LLZ}$) at different temperatures.

T, °C	σ_{LLZ}, S cm^{-1}	$\sigma_{LBV\|LLZ}$, S cm^{-1}				
		650 °C 3 min	700 °C 1.5 min	750 °C 0.5 min	800 °C 0.5 min	850 °C 0.5 min
140	$4\cdot10^{-4}$	$2\cdot10^{-7}$	$2\cdot10^{-7}$	$1\cdot10^{-6}$	$4\cdot10^{-7}$	$4\cdot10^{-7}$
170	$1\cdot10^{-3}$	$6\cdot10^{-7}$	$9\cdot10^{-7}$	$5\cdot10^{-6}$	$2\cdot10^{-6}$	$1\cdot10^{-6}$
200	$3\cdot10^{-3}$	$2\cdot10^{-6}$	$4\cdot10^{-6}$	$1\cdot10^{-5}$	$5\cdot10^{-6}$	$3\cdot10^{-6}$
230	$7\cdot10^{-3}$	$6\cdot10^{-6}$	$7\cdot10^{-6}$	$3\cdot10^{-5}$	$1\cdot10^{-6}$	$8\cdot10^{-6}$
260	$1\cdot10^{-2}$	$2\cdot10^{-5}$	$1\cdot10^{-5}$	$8\cdot10^{-5}$	$3\cdot10^{-5}$	$2\cdot10^{-5}$
290	$3\cdot10^{-2}$	$5\cdot10^{-5}$	$4\cdot10^{-5}$	$2\cdot10^{-4}$	$6\cdot10^{-5}$	$4\cdot10^{-5}$
330	$6\cdot10^{-2}$	$2\cdot10^{-4}$	$1\cdot10^{-4}$	$4\cdot10^{-4}$	$2\cdot10^{-4}$	$1\cdot10^{-4}$

>750 °C (4×10^{-7} S cm^{-1} at 140 °C) can be related to the boron-containing phase formation or the interaction of the glass with the $Li_7La_3Zr_2O_{12}$ substrate.

Then, the electrolyte resistance was deducted from the sample's total resistance in order to determine the resistance of crystallized glass and the interface between the solid electrolyte and the electrode. The temperature dependences of the total conductivity, which includes the contributions of LBV glass crystallized at various temperatures and the interface between LBV and LLZ, are shown in Fig. 5. The activation energy of the conductivity (E_a) for the samples crystallized for 3 min at 650 °C is 81 ± 2 kJ mol^{-1}. Apparently, poor contact between the cathode material and the electrolyte leads to the large value of E_a. For other samples, the activation energy decreases, and its value is about 67 ± 1 kJ mol^{-1}. It can be assumed that the deposited glass forms a better contact with the solid electrolyte at temperatures above 700 °C.

The universality of the optimal conditions of the glass deposition was demonstrated using LBV crystallization on a LAGP substrate. The $30Li_2O\cdot47.5V_2O_5\cdot22.5B_2O_3$ glass was deposited on $Li_{3.65}Al_{0.05}Ge_{0.95}P_{0.2}O_4$ solid electrolyte under various conditions: 700 °C (1.5 min), 750 °C (0.5 and 1 min), and 800 °C (0.5min). The LBV|LAGP|GaAg samples obtained after holding at 750 °C for 0.5 min possessed the highest total conductivity. Therefore, this crystallization mode of lithium borate-vanadate glass can be used for creating the optimal interface with other lithium-ion solid electrolytes.

The polarization resistance measurements of three-electrode Ni|LBV|LLZ|GaAg|Ni cells were performed by a complex of electrochemical techniques. Samples obtained at the LBV glass deposition for 1.5 min at 700 °C (sample 1) and for 0.5 min at 750 °C (sample 2) were studied. The impedance plots in Fig. 6 have two overlapping and oblate semicircles for 30–70%. Such a semicircle appearance

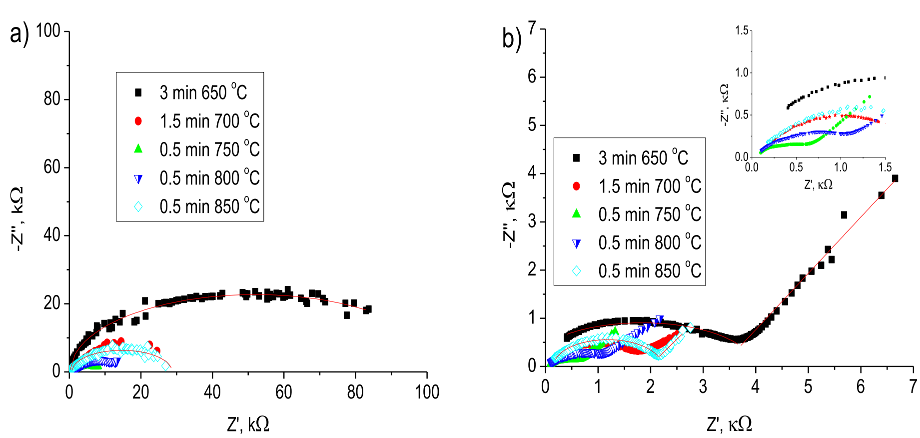

Fig. 4. Impedance plots of Ag|LBV|LLZ|GaAg|Ag cells measured at 230 (a) and 320 °C (b) with glass electrode crystallized on the LLZ substrate under various conditions (see legend).

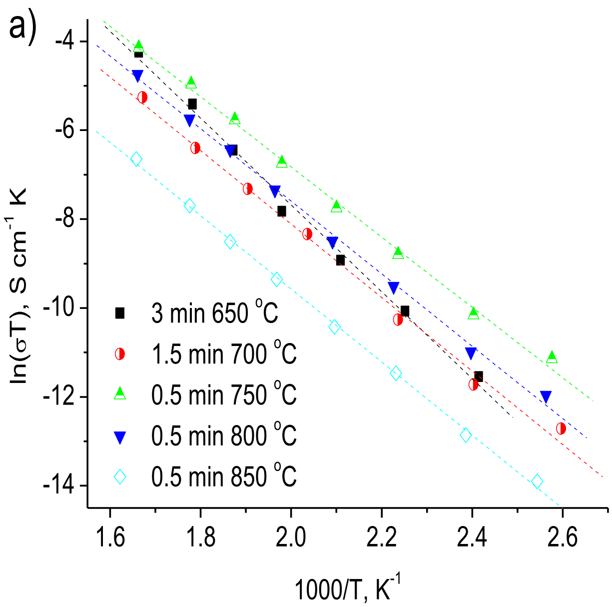

Fig. 5. Arrhenius plots for the total conductivity of the Ag|LBV|LLZ|GaAg|Ag samples obtained at different conditions of glass crystallization (see legend) without resistance of solid electrolyte.

was also observed in other works [25,27,29]. The high-frequency region was extrapolated to the intersection with the real resistance axis, the value of which coincides with the bulk resistance of the solid electrolyte at the appropriate temperature. In the simplest case, the equivalent circuit can be written as a serial connection of the ohmic resistance of the charge transfer through the material and two parallel connected blocks of the charge transfer resistance during the electrochemical reaction and a constant phase angle element (CPA), thereby characterizing the capacity at the electrode interface [48].

The potential responses for the pulsed current had a traditional trend with the initial jump corresponding to the voltage drop due to the pure charge transfer through the cell, the subsequent exponential increase of polarization, and the exit to the plateau [49]. It should be noted that the outlet to the plateau was indicated after 8–14 μs from the pulse beginning at all studied temperatures. The resistance values obtained by extrapolating the current switching curves coincided with the total resistance of the electrolyte material. The resistances calculated from the current and the plateau potential values at high temperatures (330 and 290 °C) matched with the value of R_{ct1}, which was determined from the impedance plots. Therefore, the equivalent circuit element containing R_{ct1} characterizes the electrochemical reaction course at the working electrode. This reaction can be written as:

$$kLi_2O\text{-}B_2O_3\text{-}nVO_2\text{-}mV_2O_5 - \bar{e} = xLi^+ + (k\text{-}x/2)Li_2O\text{-}B_2O_3\text{-}(n\text{-}x)VO_2\text{-}(m + x/2)V_2O_5.$$

This reaction describes the lithium deintercalation from the electrode surface to the electrolyte (to the field direction), which is attended with an increase in the vanadium oxidation degree in the glass composition. At lower temperatures, the R_{ct1} values did not coincide with the resistance obtained from the potential and the plateau current of the pulse, which indicates an increase in the relaxation time of the system.

The potentiometric studies revealed that when the polarization in the system continues to increase for a sufficiently long time in the galvanostatic mode, a plateau reaches at a current of 10 μA within 100–1000 s. It is difficult to speculate about reaching the plateau at 50 and 100 μA. Moreover, the overvoltage resistance, calculated from potentiometric curves, was within the limits of error and agrees with R_{ct2}, as determined by the extrapolation of the low-frequency arch of the impedance plots to the axis of the real resistance. This indicates that this value is the last contribution to the total resistance of the investigated cell. In our opinion, this polarization component characterizes the difficulties of the electrochemical reaction occurring on the counter electrode, which can be written as:

$$Li^+ + \bar{e} = Li^0(Ag).$$

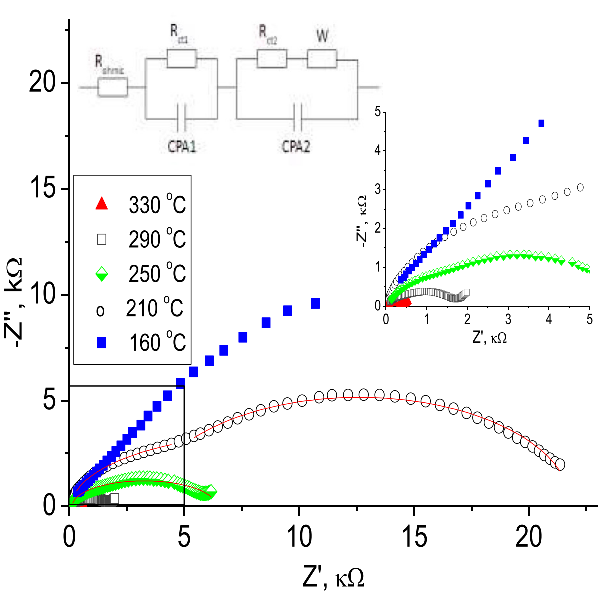

Fig. 6. Impedance plots of a Ni|LBV|LLZ|GaAg|Ni cell with LBV glass, crystallized on the LLZ substrate at 750 °C for 0.5 min and equivalent circuit.

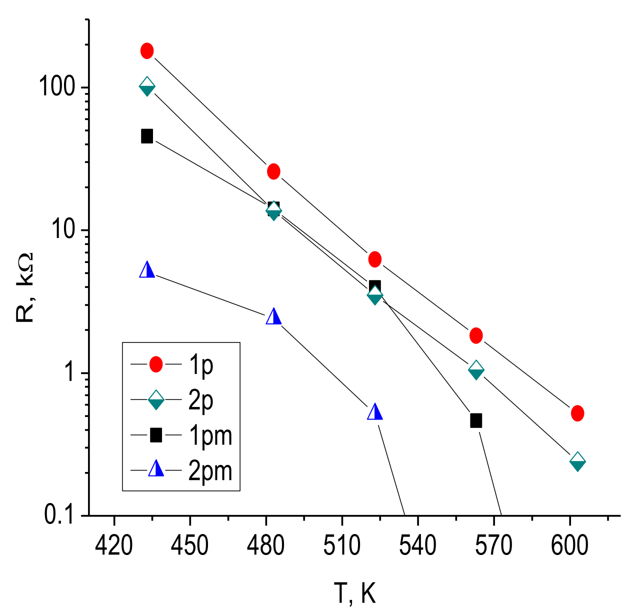

Fig. 7. Polarization resistances determined at different temperatures for the Ni|LBV|LLZ|GaAg|Ni cell with LBV glass, crystallized on the LLZ substrate at 700 °C for 1.5 min (sample 1) and 750 °C for 0.5 min (sample 2), from the pulse plateau (1p, 2p) and the plateau of the potentiometric method (1pm, 2pm).

Table 4
Polarization resistances calculated from galvanostatic plateau data for samples with LBV glass, crystallized on the LLZ substrate at 700 °C for 1.5 min (sample 1) and 750 °C for 0.5 min (sample 2): pulse measurement (R_{pulse}) and long-time potentiometric study ($R_{potentiometric}$).

T, °C	R_{pulse} (sample 1), Ω	$R_{potentiometric}$ (sample 1), Ω	R_{pulse} (sample 2), Ω	$R_{potentiometric}$ (sample 2), Ω
330	10	521	10	240
290	464	1825	12	1052
250	3971	6231	517	3512
210	14094	25736	2406	13681
160	45617	180033	5106	101868

The response of working electrode polarization is caused by the slowness of the last reaction; hence the lithium cations concentration in the electrolyte volume increases. This shifts the equilibrium of the electrode reaction on the working electrode towards the initial substances. The values of the polarization resistances described by the elements R_{ct1} and R_{ct2} at different temperatures are shown in Fig. 7. The polarization resistance calculated from pulse measurements has minimal values (around 10 Ohm) at the temperatures of 330 and 290 °C for samples 1 and 2 respectively (Table 4). Another potentiometric plateau can be found at the much longer time of the constant current exposure. The resistance values calculated from these data exceeds the resistance values calculated from pulse plateau data under the same conditions by one order of magnitude. The polarization resistance calculated from the long-time potentiometric data exceeds 100 kOhm at 160 °C and lower temperatures. Under the same conditions, both polarization resistances of sample 1 exceed the values of sample 2 in two times. The resistance values at 160 °C for sample 2 are about 5 kOhm (pulse data) and 100 kOhm (long-time potentiometric data). According to the investigated data, it can be concluded that the sample obtained for 0.5 min at 750 °C has not only lower cell resistance values but also lower polarization values. Thus, this condition can be considered optimal for applying the LBV glass electrode.

A typical polarization curve obtained from the voltammetry data is presented in Fig. 8. It can be seen that the current flow through the cell has a predominantly difficulties of activation nature. It can be concluded that the activation of the reaction occurring on the counter electrode (GaAg) mainly complicates the current flow. At the same time, the outflow of the lithium cations from the working electrode is complicated; the limiting stage is assumed to be the electrochemical reaction on the counter electrode. This statement is also supported by the fact that the total resistance of the electrolyte at corresponding temperatures is 1–3 orders of magnitude lower than the value of the total polarization resistance in the cell.

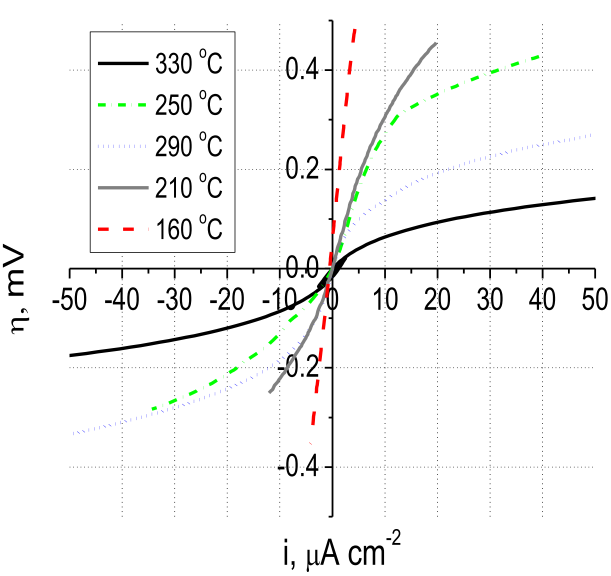

Fig. 8. Polarization curves of a Ni|LBV|LLZ|GaAg|Ni cell with LBV glass, crystallized on the LLZ substrate at 750 °C for 0.5 min.

LBV glass deposited at optimal conditions (0.5 min at 750 °C) was used in a two-probe cell under cycling. The charge-discharge curves for the LBV|LLZ|GaAg cell are depicted in Fig. 9. The initial voltage of the assembled cell was around 0.5 V, the fully-charged state had a voltage of 1.5 V, and the fully-discharged cell's voltage was around 0.5 V. The overvoltage values differed significantly for the two processes: charging with 1 μA current flow (2.6 μA cm^{-2} current density) gives an overvoltage of 0.25 V, whereas discharging with the same current gives an overvoltage of around 4 V (Fig. 9a). This fact supports the assumption that the GaAg electrode acts as the limiting factor for energy conversion in the cell. Charging at 50 μA (130 μA cm^{-2}) results in an overvoltage of around 1.5 V (Fig. 9b). It should be noted that in the case of a cycling experiment, overvoltage characterizes the charge/discharge process that combines half-reactions on both electrodes, and thus the overvoltage values are the sum of overvoltages at the LBV and GaAg electrodes. It was discussed above basing on three-probe cell polarization study, that GaAg electrode acts as limiting in all cases while LBV electrode shows satisfactory characteristics for being applied as cathode for lithium-ionic or lithium power sources.

3. SEM investigation of the LLZ|LBV interface

Scanning electron microscopy (SEM) study of the sample cross-section was carried out to determine the thickness of the cathode coating, the depth of glass penetration into the solid electrolyte, as well as the presence of an interaction between the glass and electrolyte ceramic. The cross-section of $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ glass on the substrate of $Li_7La_3Zr_2O_{12}$ solid electrolyte, obtained under various temperature and time conditions, is shown in Fig. 10. Three regions can be separated for all samples: the first upper layer of crystallized glass (thickness of 20–35 μm), the second layer (38–75 μm) is a mixture of crystallized glass and ceramic grains, and the third is the LLZ ceramic.

The glass obtained for 3 min at 650 °C crystallizes in the form of a needle structure and has a very uneven surface with various protrusions (Fig. 10a). The average thickness of this layer is about 35 μm (Fig. 10b). Under these conditions, the glass penetrates into the ceramic substrate by 38 μm, without visual interaction, but the contact between the cathode and the electrolyte is weak. Apparently, such a structure and lose contact lead to the low conductivity values of this coating and large values of the activation energy (see Part 2).

LBV evenly covers the ceramic surface when the temperature increases up to 750 °C. The layer thickness has a uniform structure, with a thickness of ~20 μm (Fig. 10c and d), and the glass deeply penetrates into the ceramics (by about 70–75 μm). It is clearly seen from the micrographs (Fig. 10c) that the glass fills the pores in the

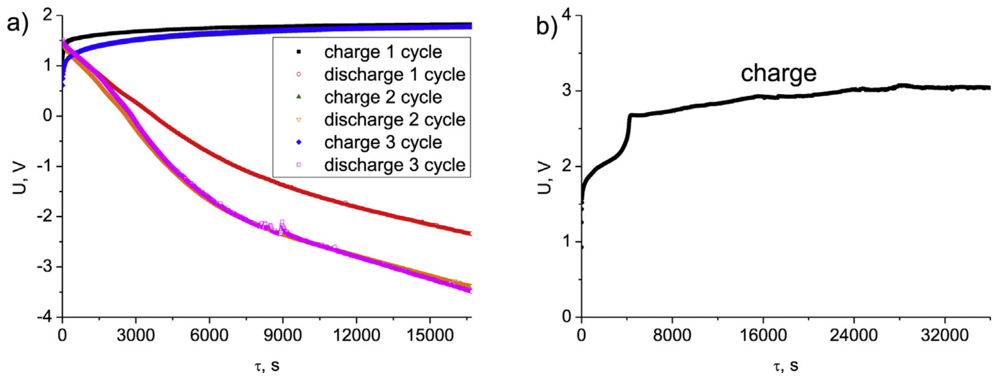

Fig. 9. Charge-discharge curves of the LBV|LLZ|GaAg cell with LBV glass crystallized on the LLZ substrate at 750 °C for 0.5 min. LBV glass was set as the working electrode (+) under a current of 1 (a) or 50 (b) μA, (2.6 and 130 μA cm^{-2}, correspondingly).

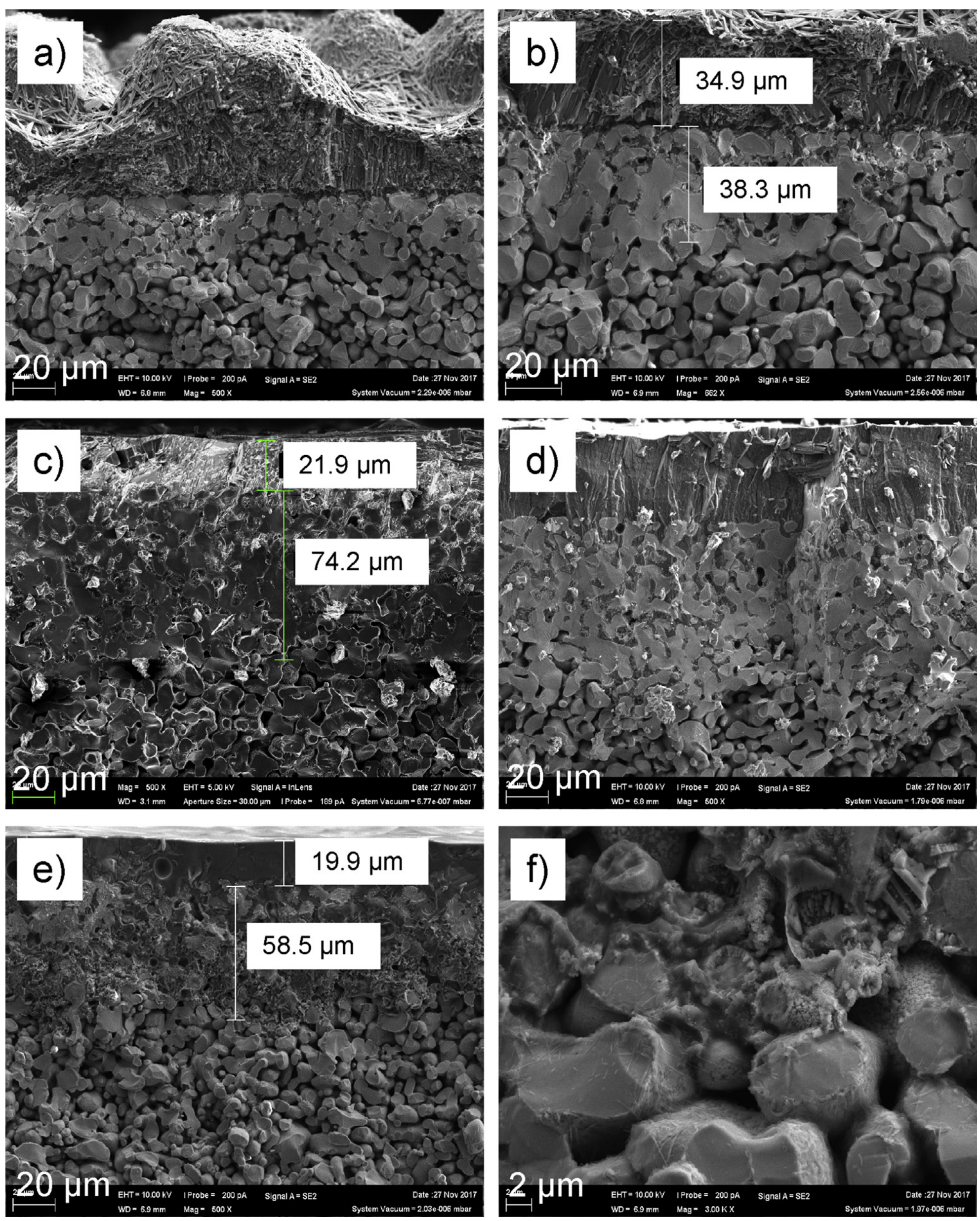

Fig. 10. SEM micrographs of the cross-section of LBV glass, crystallized on the LLZ substrate under various conditions: 650 °C for 3 min (a,b), 750 °C for 0.5 min (c,d), and 850 °C for 0.5 min (e,f).

solid electrolyte, covers the grains of the ceramics, and provides a tight contact between them. The areas of the ceramic particles and crystallized glass are clearly separated. Therefore, one can conclude that there is no interaction between these two phases. This provides high electrical conductivity values and low polarization resistance of samples.

The glass also forms an even layer of about 20 μm (Fig. 10e and f) on the substrate surface at 850 °C, but there is an obvious interaction with the ceramics. It is clearly shown that the boundaries between ceramic grains are being destroyed; the thickness of this second layer is about 58 μm. The process of interaction between the glass and ceramics, as well as the formation of the boron-containing phases established by the XRD, has a negative effect on the electrical conductivity of the samples and leads to its decrease in the studied temperature range.

Thus, it is important to obtain a certain phase composition during the partial or complete glass crystallization and its further use as a cathode material, as well as to provide close contact between the applied cathode and the electrolyte substrate without interaction between them. By XRD and XPS, it was established that bronzes of LiV_2O_5 and LiV_3O_8 vanadium formed at 650 °C (3 min) are able to intercalate and deintercalate lithium ions. But the resulting cathode layer, according to the electron spectroscopy data, has poor contact with the LLZ substrate and, as a consequence, the cell has a low conductivity and high activation energy values. The temperature rise to 700 °C did not result in an increase in the electrical conductivity. According to the XRD and XPS, the lithium vanadates $LiVO_3$ and Li_3VO_4 are formed, which contain vanadium only in the 5^+ oxidation state and cannot be used as electrode material. LBV glass crystallized for 0.5 min at 750 °C has the highest electrical conductivity, consists of the vanadium bronzes, and provides a good and tight contact between the electrode and the electrolyte. The further increase in temperature leads to the decrease in the electrical conductivity of samples, the formation of boron-containing phases, and the glass and LLZ interaction.

4. Conclusions

The $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ glass was deposited on $Li_7La_3Zr_2O_{12}$ solid electrolyte under various conditions: 650 °C (for 3, 4 and 5 min), 700 °C (for 1.5, 2 and 3 min), 750 °C (0.5, 1 and 2 min), and also at 800 and 850 °C for 0.5, 1 and 1.5 min. It was established by XRD that the LBV glass crystallized with the formation of various vanadium phases – $LiVO_3$, Li_3VO_4, LiV_2O_5, and LiV_3O_8 at 650, 700, and 750 °C. The $Li_2B_4O_7$ and B_2O_3 boron-containing phases also appeared at higher temperatures, and their proportion increased with increasing holding time. According to the XRD and XPS analysis, the vanadium bronzes (LiV_2O_5 and LiV_3O_8) with vanadium in the 4^+ and 5^+ oxidation states were formed under glass crystallization for 0.5 min at 750 °C. These bronzes are well known as cathodes in lithium-ion batteries. According to SEM investigation, the LBV glass, which was crystallized at 750 °C, creates the optimal interface and good contact between electrode and electrolyte without interaction between them. At low temperatures, the cathode coating was uneven, and a good contact was not achieved. The interaction between the cathode material and the ceramic electrolyte was observed at temperatures above 750 °C. Therefore, a temperature of less than 800 °C should be used for LBV glass deposition.

The resistance of the Ag|LBV|LLZ|GaAg|Ag cells was measured by the impedance spectroscopy, and it was shown that the highest conductivity at each crystallization temperature is achieved at the minimal holding time. The samples annealing for 0.5 min at 750 °C had the highest total conductivity – 1×10^{-6} and 4×10^{-4} S cm^{-1} at 140 and 330 °C, respectively. According to the electrochemical study of Ni|LBV|LLZ|GaAg|Ni cells, it can be concluded that the sample annealed for 0.5 min at 750 °C has not only lower cell resistance values but also lower polarization values. The LBV|LLZ|GaAg cells were cycled with edge values of voltage of 1.5 V and 0.5 V (LBV electrode was set as (+)) with overvoltage values of 0.25 V at 2.6 μA cm^{-2} and 1.5 V at 130 μA cm^{-2} under charging. The universality of this crystallization mode was demonstrated using, as an example, another lithium-solid electrolyte (LAGP). Thus, the annealing for 0.5 min at 750 °C is the optimum regime for the $30Li_2O \cdot 47.5V_2O_5 \cdot 22.5B_2O_3$ glassy cathode deposition on the substrate of lithium-ion solid electrolyte when producing middle-temperature all-solid-state batteries.

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